# Online Resource 1 for "Silent but deadly? Physics says it depends, memory says mostly"

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## S1 Odorant chemistry

### S1.1 Methods

*Odour activity.* Each sulfur gas contributes an odour activity value (OAV), its mole fraction over its detection threshold, $\mathrm{OAV}_i = x_i/\theta_i$. The passage's OAV is their sum; we assume additivity and do not model mixture suppression. We compare two measured compositions:

- 87 passages of 16 volunteers [S1], used throughout the dispersion model;

- 28 passages measured by gas chromatography: $H_2S$ 149, methanethiol 216 and dimethyl sulfide 144 nmol $L^{-1}$ [S2].

Three threshold sets bracket the observer: a trained panel [S3]; population values, the geometric means of detection thresholds for $H_2S$ and methanethiol [S4] and, for dimethyl sulfide, the median recognition threshold of an untrained panel of 35 [S5]; and the lowest values in Verschueren's handbook, as tabulated in ref. S2.

*Removal check.* Zinc acetate precipitates $H_2S$ and thiols; activated charcoal adsorbs all three gases. For both treatments, ref. S1 reports the gas concentrations and two judges' ratings before and after. We predict the rating ratio as $(\mathrm{OAV_{after}}/\mathrm{OAV_{before}})^n$ with $n = 0.20$–$0.45$, an assumed bracket around the 0.28 measured for canine flatus [S6], without fitting.

*Buoyancy.* The reduced gravity $g' = g(\rho_a - \rho_g)/\rho_a$ of each of 16 measured compositions [S7] is split into a thermal part, $g(1-T_a/T_g)$, for gas at 37 °C in air at 22 °C, and a compositional part, $g(M_a - M_g)/M_a$, at equal temperature with air at 50% relative humidity. The water-vapour content of flatus has not been measured, so we bracket dry and saturated gas.

*Gas-phase and surface losses.* Gas-phase losses use IUPAC rate constants at 298 K for reaction with OH, $O_3$, $NO_3$ and Cl [S8], with oxidant levels measured in an office: OH 0.6–9.8 × $10^5$ $cm^{-3}$ and $O_3$ 0.7–2.6 ppb (daily means) [S9]. Two assumed upper cases add OH $10^6$ $cm^{-3}$ with $O_3$ 40 ppb, and Cl $10^6$ $cm^{-3}$ during hypochlorite cleaning. No deposition velocity $v_d$ has been published for these gases on indoor surfaces, so surface loss is bracketed as $k = v_d\, S/V$ with $S/V = 3\ \mathrm{m^{-1}}$.

*Dilution in a panel test.* Ref. S2 reports the lowest source concentration that every panel member detected when 100 mL was released below the nose. Divided by a detection threshold, it gives the dilution between release and nose in that test, which we compare with the breath-averaged dilution of the analytic model.

## S1.2 Results

*Which molecule.* Over the two measured compositions and three threshold sets, methanethiol carries 43–94% of the summed OAV and never less than $H_2S$ (Figure S1a).

- With the composition of ref. S1, methanethiol and $H_2S$ carry about equal shares (43–54% and 43–46% with trained-panel and population thresholds). Methanethiol is one fifth as concentrated, but its threshold is 5–6 times lower.

- With the composition of ref. S2, or with the lowest reported thresholds, methanethiol carries 60–94%.

The two datasets disagree on the concentrations themselves: $H_2S$ is five times methanethiol in one and slightly below it in the other, a difference ref. S2 attributes to the bean-rich diet of the first. In that study's panel test, gas mixtures with methanethiol above its flatus threshold smelled like most passages.

Composition does not enter the silent/audible comparison: transport is linear and acts on all three gases alike, so every ratio is the same for any composition. Absolute OAVs are 11–38% lower with the second composition.

*The intensity law with real flatus.* Zinc acetate lowers the summed OAV by a factor of 14. Stevens' law then predicts rating ratios of 0.31–0.59, and the two judges gave 0.28 and 0.59 [S1]. Activated charcoal lowers the OAV about 260-fold and is rated far weaker than the power law predicts, as power laws fail near threshold. The silent/audible comparison uses supra-threshold ratios, where the law held.

*What makes a passage rise or sink.* Composition sets most of the buoyancy (Figure S1b).

- For the median passage the compositional part of the reduced gravity (+1.25 $\mathrm{m\,s^{-2}}$ dry, +1.40 saturated) is about three times the thermal part (+0.47).

Table S1: Gas-phase lifetime of the odorants and the fraction lost in 120 s. Office oxidant levels are measured [S9]: low, OH $0.6 \times 10^5$ cm$^{-3}$ and $O_3$ 0.7 ppb; high, OH $9.8 \times 10^5$ cm$^{-3}$ and $O_3$ 2.6 ppb (the highest daily means of each, measured on different days). The cleaning case assumes Cl $10^6$ cm$^{-3}$. Rate constants from ref. S8.

| scenario | $H_2S$ | $CH_3SH$ | DMS |
|---|---|---|---|
| office, low | 40 d / 0.003% | 5.7 d / 0.03% | 28 d / 0.005% |
| office, high | 2.5 d / 0.06% | 8.6 h / 0.4% | 1.8 d / 0.08% |
| hypochlorite cleaning | 3.5 h / 0.9% | 1.2 h / 2.8% | 0.8 h / 4.0% |

- $H_2$ and $CH_4$ make flatus light and $CO_2$ makes it heavy; the compositional part falls with $CO_2$ content ($r = -0.80$).
- 6–12% of passages sink even while hot. After cooling to room temperature, 25–38% are heavier than air.

The dispersion model uses these signed values of $g'$.

*Losses on the way to the nose* (Table S1).

- In an office, gas-phase oxidation removes at most 0.4% of any of the three gases in 120 s, and hypochlorite cleaning less than 4%.
- Surface deposition is unconstrained by data. It removes 0.7% in 120 s at $v_d = 0.002$ cm s$^{-1}$ and 51% at 0.2 cm s$^{-1}$. Treating the odorants as passive scalars therefore holds against chemistry by two to three orders of magnitude, and against deposition only if $v_d \lesssim 0.01$ cm s$^{-1}$.
- Deposition would remove most from the silent release, which stays close to the body and the seat, and would lower its doses further in the simulated scenes.

*How dilute a passage arrives.* The panel test of ref. S2 implies a dilution of 0.6–13 $\times 10^4$ between release and nose, depending on the threshold set. The breath-averaged dilution of the analytic model at the neighbour has a median of $1.6 \times 10^3$ (10–90%: $2.6 \times 10^2$–$2.7 \times 10^4$). The panel value is biased upwards, because it requires the least sensitive panellist to detect but divides by the thresholds of the most sensitive. The analytic model is at the concentrated end of the panel range, as intended for an upper bound, and within about an order of magnitude of it.

*Composition and volume.* Larger passages contain more fermentation gas and less nitrogen (per passage, $r = +0.56$ for $CO_2$ and $-0.50$ for $N_2$, $n = 87$), while the sulfur concentrations do not change with volume ($|r| \leq 0.22$, not significant); sulfur mass therefore rises with passage volume [S7]. At high production rates almost all flatus is fermentation gas [S10]. Under bean, fibre, fruit and lactulose challenges the fermentation-gas fraction rises with output ($r = +0.86$ to $+0.98$ across diet or treatment means, computed from the published tables [S11–S16], for ref. S13 from the 12 control and bean rows without silicone; in one person from 26% at 55 mL to 79% at 720 mL per 4 h [S17]). One study is an exception: on an ordinary diet the fraction did not rise with output ($r = -0.41$ across 20 subjects), and after brussels sprouts the extra gas was mostly nitrogen [S18]. The density of the gas does not follow the amount: carbon dioxide and hydrogen rise together, and the mean molar mass, 22–32 g mol$^{-1}$ across diets, is set by the substrate's $CO_2$:$H_2$ ratio (its correlation with output ranges from $-0.37$ to $+0.90$ across seven studies). Composition therefore cannot link audibility to the amount of gas, since it enters audibility only through density (Section S2).

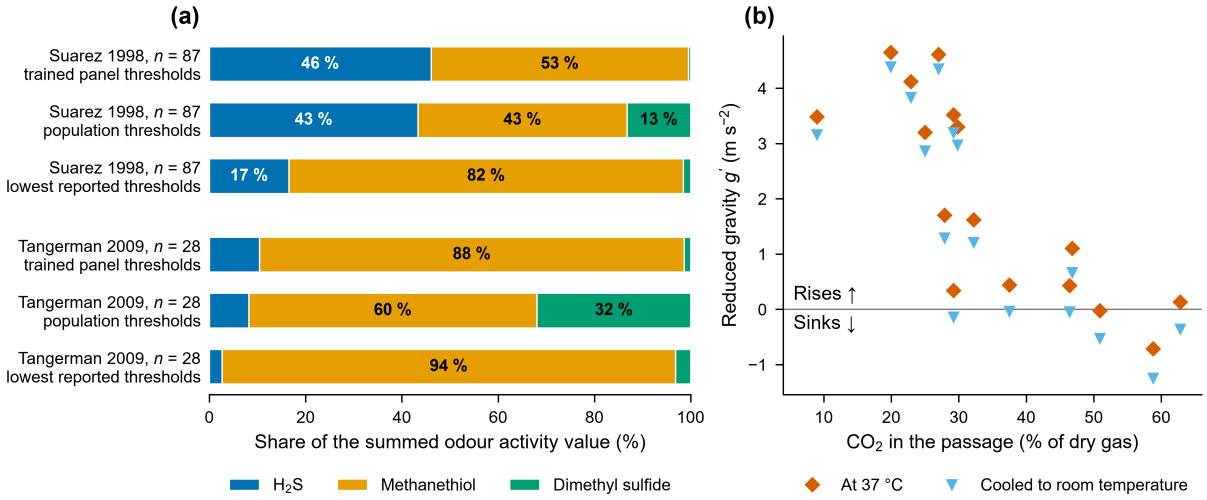



Figure S1: Odorants and buoyancy. (a) Share of the summed OAV carried by $H_2S$, methanethiol and dimethyl sulfide, for two measured compositions [S1, S2] under three threshold sets. (b) Reduced gravity of 16 measured compositions [S7] against their $CO_2$ content, at 37 °C (dry basis; water saturation shifts these values by $-0.04$ to $+0.29\,\mathrm{m\,s^{-2}}$) and after cooling to room temperature.

## S2 Acoustics

*Duration and orifice.* Flatus passes during an anal relaxation lasting $19 \pm 6$ s [S19] (relaxations accompanying pan-colonic pressurizations last $23 \pm 4$ s [S20]), which bounds the duration $T$ from above, and the mid anal canal widens to about 8 mm only at balloon pressures (40–55 mmHg) [S21, S22] above the rectal pressures at which flatus usually passes (median rise 8 mmHg, range $-10$ to 88) [S23], which bounds the effective diameter $d$.

*Valve.* The anal canal is modelled as a lip reed, a soft-tissue valve that self-oscillates in the flow through it, as the lips do on a brass instrument. It oscillates when the driving pressure $\frac{1}{2}\rho(u_0/C_\mathrm{d})^2$ exceeds a threshold $P_\mathrm{th} = 0.6$–$2.5$ kPa (nominal 1.2 kPa), with discharge coefficient $C_\mathrm{d} = 0.85$. The closest measured soft-tissue sphincter source, the pharyngo-oesophageal segment that drives oesophageal and tracheo-oesophageal speech, phonates at 1.1–2.5 kPa: sustained phonation from 1.1 kPa [S24], 1.8 kPa below the segment at minimum loudness [S25] and about 2.5 kPa on average during speech (individuals 1.0–3.0 kPa) [S26]. This bounds the onset from above; lips on a brass instrument start at about 0.6 kPa [S27]. This gives an onset velocity $u_\mathrm{onset} \approx 42\,\mathrm{m\,s^{-1}}$ (21–71).

An oscillating valve radiates as a harmonic monopole with harmonics falling by 12 dB per octave. Its fundamental is taken as 160 Hz (60–300 Hz), between oesophageal and tracheo-oesophageal speech at 60–150 Hz [S24, S28–S30] and recordings of flatus near 270 Hz, which are not peer-reviewed [S31]. A release counts as *audible* when at least one harmonic exceeds both the ISO 226 hearing threshold [S32] and the masked threshold of the background, computed from the critical ratio [S33]. Backgrounds range from a 35 dB(A) living room to a 77 dB(A) subway car.

*Valve state.* Manometry shows flatus passing through a relaxed sphincter, at a median rectal pressure rise of 8 mmHg (range $-10$ to 88; 0–15 mmHg when flatus is only sensed) [S23], with a fall in anal pressure [S19], but also against a contracted sphincter [S34]. The 85% silent share treats the onset pressure as a property of the tissue and does not distinguish the two.

## S3 Proofs of Propositions 1–3

Equation numbers refer to the main text.

*Proposition 1.* (i) follows from integrating Eq. (2), $C = G{*}q$, over $t$. For (ii), $C_\mathrm{s} = G{*}q_\mathrm{a}{*}\kappa = C_\mathrm{a} * \kappa$, and Young's inequality gives $\|C_\mathrm{s}\|_\infty \le \|C_\mathrm{a}\|_\infty \, \|\kappa\|_1 = \|C_\mathrm{a}\|_\infty$. □

A top-hat release lasting $nT$ is a top-hat lasting $T$ convolved with $\kappa = n^{-1} \sum_{j=0}^{n-1} \delta(t - jT)$, so stretching a release by an integer factor is covered. The breath-averaged OAV of Eq. (1) is itself a convolution of $C$ with a non-negative kernel, so the same argument bounds it.

*Proposition 2.* Let the release last $T_\mathrm{r}$, so that $q$ vanishes outside $[0, T_\mathrm{r}]$. Then $C(t) = \int_0^{T_\mathrm{r}} G(t-s)\, q(s)\, \mathrm{d}s$, and as $t \to \infty$, $G(t-s) \to 1/V_\mathrm{room}$ uniformly for $s \in [0, T_\mathrm{r}]$, so $C(t) \to V_\mathrm{room}^{-1} \int q \, \mathrm{d}s = V/V_\mathrm{room}$ for any $q$. □

*Proposition 3.* On $\neg H$, noticing requires smelling, so $\neg H \cap N = \neg H \cap S$ and $P(S \mid \neg H, N) = 1$. On $H$, $N$ is certain, so conditioning on $N$ adds nothing. □

## S4 Near field of the analytic model

The release behaves as a starting jet. Its tip follows $x = \Gamma(\dot{Q}_0 u_0)^{1/4} t^{1/2}$, where $\dot{Q}_0 = V/T$ is the volume flow rate and $\dot{Q}_0 u_0 = M_0$ the kinematic momentum flux, and once the source stops it follows the puff law $x^4 = C_p^4 V u_0 t$, with $\Gamma = 2.8$ and $C_p = 2.6$ from the corrected analysis [S35] of the measurements of ref. S36. The centreline dilution is $x/(K_c d^*)$ with $K_c = 5.11$ [S37], where $d^* = d\,(\rho_0/\rho_\mathrm{a})^{1/2}$ is the momentum diameter of a release of density $\rho_0$ into air of density $\rho_\mathrm{a}$. In parallel, a buoyant thermal grows with entrainment coefficient $\alpha = 0.25$ [S38], and the larger of the two dilutions is kept. The near field ends when the self-induced velocity has fallen to the ambient velocity $u_h$ (0.05–0.25 m s$^{-1}$).

For a steady jet, a starting jet and a puff alike, the momentum carried by the cloud is conserved while it entrains, so the entrained mass grows as the inverse of the remaining velocity. At the hand-off velocity $u_h$ the dilution is therefore proportional to $u_0/u_h$, which is Eq. (3); the prefactor $\beta$ collects the jet or puff constants above and is of order one.

## S5 CFD validation

- *Free round jet* (axisymmetric, Re = 6700, three meshes). The velocity-decay constant is 5.0 and the scalar constant $K_c = 4.3$–$4.6$, with a grid uncertainty of at most 3.7%. The $k$–$\omega$ SST model spreads round jets about 25% too fast, which over-dilutes audible releases by about 15% and so favours the silent release.

- *Convective boundary layer* of a simplified seated manikin with skin at 33 °C, compared with thermal-manikin data [S39]. The face-front peak velocity is 40–56% too high at room temperatures of 20–26 °C; its change with room temperature is reproduced (−27% against −33% measured).

- *Near-body source to breathing zone.* The chamber of ref. S40 (4.6 × 4.6 × 2.36 m, 115 m$^3$ h$^{-1}$ through two ceiling supplies, exhaust at 1.8 m) with the scene manikin, chair, skin temperature and solver; a 3.5 L min$^{-1}$ source between the thighs at 0.5 m runs for 10 min. Intake fraction $\mathrm{iF} = Q_\mathrm{b} \int C_\mathrm{bz}\, \mathrm{d}t/m$ with $Q_\mathrm{b} = 0.5$ m$^3$ h$^{-1}$, from the breathing-zone probe at 1.1 m; after 22 min the room-mean level decays at the measured air-exchange rate. Result 8.1, 9.1 and 11.0 per mille over 10, 22 and 100 min against 5.5, 7.7 and 9.8 measured; relative to the well-mixed room 11.0 and 2.6 against 7.4 and 2.3. The manikin's convective output is 79 W. Supply direction, supply temperature and exhaust size are not reported and were assumed (downward, room temperature, 0.3 m square).

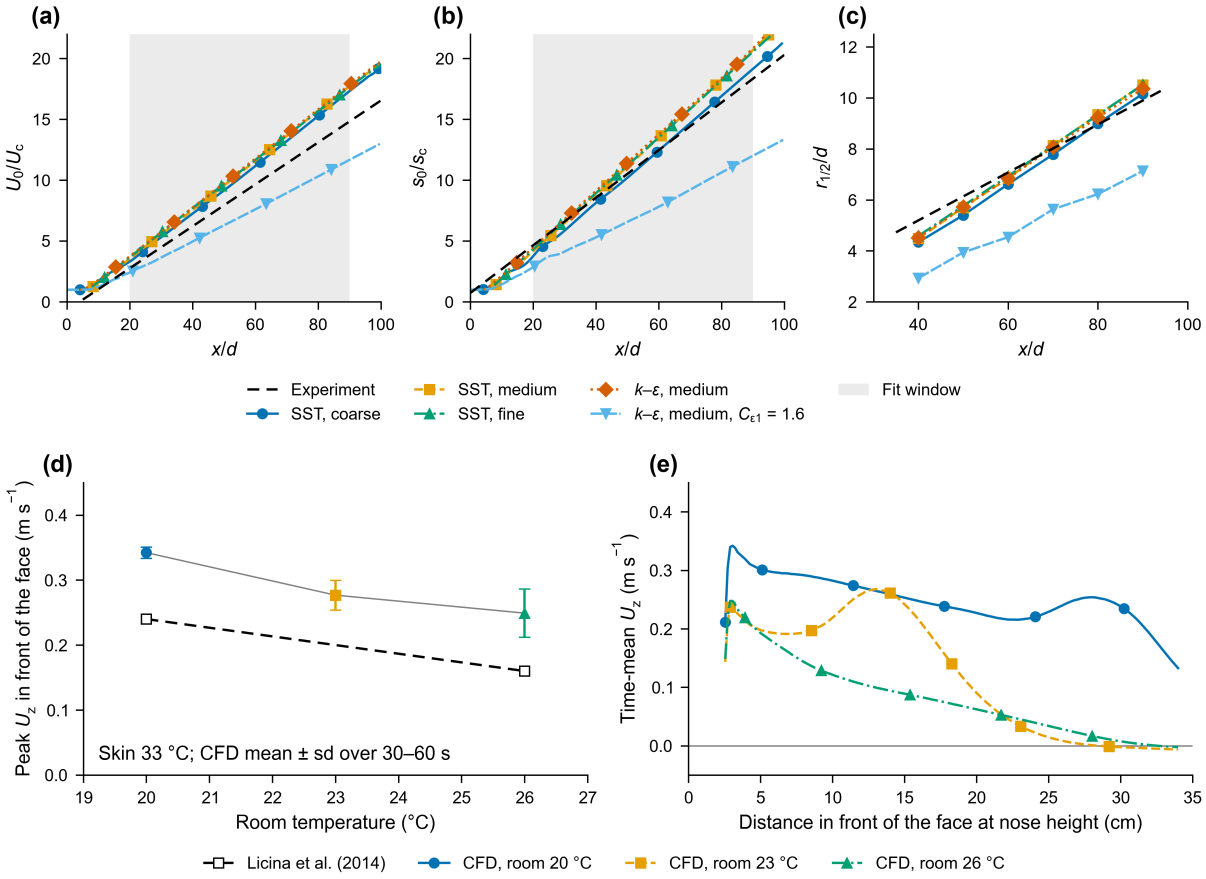



Figure S2: Validation. Top: free round jet at Re = 6700, $k$–$\omega$ SST on three meshes and two $k$–$\varepsilon$ variants, against the experimental correlations: (a) centreline velocity decay, (b) centreline scalar decay, (c) half-width (the experiment gives only the spreading rate, 0.094, drawn through the mean of the fine-mesh points); grey: the fit window. Bottom: (d) face-front peak upward velocity of the seated manikin against thermal-manikin data [S39] at 20–26 °C, (e) time-mean vertical velocity profiles at nose height.

- *Mesh.* The median silent release alone on the sofa mesh and on a mesh refined by 1.5 in every direction (0.72 and 2.2 million cells, same refinement levels): 150-s worst breath 0.42 and 1.28 at the releaser, 0.17 and 0.094 at the neighbour. Released fourth in the four-release run on the base mesh, the same passage gives 0.33 and 0.071, so the release order matters about as much as the mesh.
- *Well-mixed tail.* Two coupled 10-minute runs of the sofa (one silent, one audible 92-mL release): the tail predicts 1.00 and 0.67 (silent: releaser, neighbour) and 0.84 and 0.71 (audible) of the actual worst breath at 600 s. The neighbours' level keeps creeping up after 300 s; nothing worse arrives later.

## S6 CFD scenes

All scenes use the solver of the main text, skin at 33 °C and walls at 23 °C. Each runs 90 s without a release; the four representative releases then follow 30 s apart (strongest first) on separate scalars, each until 150 s after its start. A release enters behind the releaser's buttocks, pointing down and back, through a 1 cm-radius inlet (area $A_e$) at its own momentum flux $M_0$, and so at the velocity $u_e = (M_0/A_e)^{1/2}$ instead of $u_0$. Room air pre-dilutes it by $u_0/u_e = 2r_e/d$

Table S2: Representative releases: median and 90th percentile of the exit momentum flux within each class of the source model without the driving-pressure limit. With the limit, A50 lies at the 64th percentile of the audible passages and A90 above almost all of them, so the two audible levels bracket the limited stream.

| release | class | $V$ (mL) | $T$ (s) | $d$ (mm) | $u_0$ ($\mathrm{m\,s^{-1}}$) | $M_0$ ($\mathrm{m^4\,s^{-2}}$) | $u_e$ ($\mathrm{m\,s^{-1}}$) | pre-dilution |
|---|---|---|---|---|---|---|---|---|
| S50 | silent | 122 | 4.76 | 3.06 | 3.5 | $8.9 \times 10^{-5}$ | 0.53 | 6.5 |
| S90 | silent | 66 | 0.65 | 2.45 | 22 | $2.2 \times 10^{-3}$ | 2.65 | 8.2 |
| A50 | audible | 86 | 0.42 | 1.67 | 94 | $1.9 \times 10^{-2}$ | 7.8 | 12.0 |
| A90 | audible | 279 | 0.34 | 2.54 | 162 | 0.13 | 20.6 | 7.9 |

(Table S2), and the odorant concentration and the equivalent temperature excess of the flatus buoyancy ($g' = 1.70\,\mathrm{m\,s^{-2}}$) are reduced by the same factor, so the momentum, odorant and buoyancy fluxes are kept and the volume flux is not. A jet conserves its momentum while it entrains, so its dilution when it has slowed to the ambient velocity $u_h$ is about $u_0/u_h$ either way (Section S4).

- *Sofa.* A $3.2 \times 3.2 \times 2.6$ m living room with a 0.5 m ceiling vent for pressure relief (size and position assumed). Two seated manikins sit 0.6 m apart on a two-seat sofa whose base is solid to the floor. Mesh 0.72 million cells, 1 cm on the bodies and 2.5 mm on the inlet face. Two variants rotate the releaser's upper body about the hip: reclined by 4.5 °, so that the back closes onto the backrest, and leaning forward by 10 °, so that the gap behind the back opens.

- *Lift.* The car of a 630 kg, eight-person lift, 1.10 m wide, 1.40 m deep and 2.10 m high. Four standing manikins (nose at 1.59 m) stand in a $2 \times 2$ grid facing the door; the fan is off, and a 0.2 m ceiling opening relieves pressure (size assumed). Mesh 0.91 million cells. The releaser stands in the front row.

- *Standing pair.* The sofa room without the sofa, with two standing manikins face to face, noses 1.0 m apart. Ventilation is one square ceiling diffuser above the pair supplying $21.2\,\mathrm{L\,s^{-1}}$, 2.9 air changes per hour (EN 16798-1 Category II for two people in a low-polluting $10\,\mathrm{m^2}$ room [S41]), leaving along the ceiling at an assumed $1.5\,\mathrm{m\,s^{-1}}$ and exiting through a ceiling grille.

- *Trousers.* A porous shell 1 cm thick, 2 cm from the skin over pelvis and thighs, open at the waist and hems, with a Darcy resistance from the air permeability of denim ($0.03\,\mathrm{m\,s^{-1}}$ at 100 Pa, EN ISO 9237).

- *Large-eddy simulation.* The upright sofa, bare and in trousers, on the RANS meshes unchanged (0.72 and 1.3 million cells): WALE model with a cube-root-volume filter width, LUST convection of momentum, a Spalding wall function and a maximum Courant number of 0.8 (time step about 7 ms between releases bare and 4.5 ms in trousers, down to 30 µs during the fastest release); 60 s of spin-up from the 90-s RANS flow, then six releases 30 s apart (S50, A50, S90, A90 and the two medians again), each followed for 90 s. The mesh was not designed for wall-resolved LES, so the large-eddy runs give the spread of a breath about the mean, not a resolved boundary layer.

## S7 Scene doses

Table S3 lists the 150-s worst-breath OAV of every representative release at every nose (volumes as emitted, RANS). No silent release reaches a sofa or standing neighbour above the trained-panel threshold; in the lift every nose passes it, the person behind the releaser most of all.

Table S3: Worst-breath OAV over 150 s (trained-panel thresholds, volumes as emitted, mean flow).

| scene | nose | S50 | S90 | A50 | A90 |
|---|---|---|---|---|---|
| sofa, upright | releaser | 0.33 | 6.6 | 6.3 | 49 |
| | neighbour | 0.071 | 0.090 | 0.16 | 17 |
| sofa, reclined 4.5 ° | releaser | 0.26 | 20 | 2.1 | 25 |
| | neighbour | 0.12 | 0.95 | 0.95 | 19 |
| sofa, forward 10 ° | releaser | 0.49 | 1.2 | 0.51 | 22 |
| | neighbour | 0.30 | 0.26 | 0.29 | 36 |
| sofa, upright, trousers | releaser | 5.0 | 1.0 | 64 | 110 |
| | neighbour | 0.11 | 0.070 | 1.0 | 43 |
| standing, still | releaser | 0.13 | 0.85 | 0.26 | 1.9 |
| | neighbour | 0.0056 | 0.0016 | 0.0004 | 0.0022 |
| standing, still, trousers | releaser | 0.042 | 0.070 | 13 | 40 |
| | neighbour | 0.014 | 0.014 | 0.0033 | 0.017 |
| standing, ventilated | releaser | 0.38 | 2.8 | 15 | 6.9 |
| | neighbour | 0.17 | 0.049 | 0.018 | 0.034 |
| standing, ventilated, trousers | releaser | 0.16 | 22 | 39 | 20 |
| | neighbour | 0.15 | 0.066 | 0.099 | 0.56 |
| lift | releaser | 7.3 | 2.0 | 2.9 | 15 |
| | beside | 2.3 | 1.2 | 2.1 | 7.3 |
| | behind | 260 | 26 | 34 | 36 |
| | diagonal behind | 6.5 | 2.8 | 4.8 | 22 |

## S8 Scene weights

Weights are the hours per day an adult spends close to another person in each setting, with passages spread uniformly over waking hours: sofa 1.4 h (television with others in the room, from the American Time Use Survey; bounds 0.8–2.0), lift 0.01 h (four shared rides of about 24 s on working days; 0.0005–0.1) and standing conversation 0.5 h (0.2–1.5, an assumption from the talking and standing shares of the day), i.e. 0.73, 0.005 and 0.26. The home-centred bound takes the high sofa and low lift and standing values, the office-centred bound the reverse. Sources and derivation are in the repository (`refs/scene_weights_sources.md`). Within a scene, postures and ventilation variants count equally.

## S9 Robustness of the headline

The headline compares intensity: among pairs of one silent and one audible passage that a bystander both smells, the share in which the silent one is stronger (above the just-noticeable ratio) and the share in which the audible one is. Bystanders, bare, attentive observer, with the 88-mmHg driving-pressure limit; Table S4 gives silent stronger / audible stronger (%), the rest being ties. The pairs in which both are smelled are rare on the sofa ($3.3 \times 10^{-4}$ of pairs, still about 900 silent and 650 audible passages in effect) and common for the person behind the releaser in the lift (0.39), so a day average follows the lift unless the sofa's share of such pairs grows. Scripts: `uq/comparison_criteria.py` and `uq/intensity_robustness.py`.

- *Lift by nose:* the person directly behind the releaser 55 / 33 (both smelled in 39% of pairs); beside 31 / 56 and diagonally behind 33 / 55 (0.5 and 2.6%). The lift column is carried by the person behind.

Table S4: Silent stronger / audible stronger (%) among pairs in which a bystander smells both, per setting and for the day weightings (day: central weights; home- and office-centred days; equal weights per setting). Peak factors are the large-eddy over RANS worst breath, by class (median 2.2 silent and 1.3 audible; adverse 6.6 and 0.27); mesh is the refinement factor 3.0 of the releaser's nose on every silent dose; release order is the factor 2.4 by which a silent release alone exceeds its dose inside a multi-release run.

| variant | sofa | lift | day | home day | office day | equal |
|---|---|---|---|---|---|---|
| headline | 30 / 59 | 54 / 35 | 48 / 41 | 32 / 57 | 53 / 35 | 54 / 35 |
| mesh | 42 / 46 | 71 / 20 | 58 / 32 | 44 / 45 | 70 / 21 | 71 / 20 |
| peak factors, median | 36 / 52 | 62 / 28 | 51 / 38 | 37 / 51 | 61 / 29 | 62 / 28 |
| peak factors, adverse | 65 / 26 | 93 / 4 | 81 / 13 | 66 / 25 | 92 / 5 | 93 / 4 |
| mesh × median peak factors | 49 / 40 | 77 / 15 | 59 / 31 | 49 / 39 | 75 / 17 | 77 / 15 |
| release order | 40 / 48 | 68 / 22 | 56 / 33 | 41 / 47 | 67 / 23 | 68 / 23 |
| release order × median peak factors | 46 / 42 | 75 / 17 | 58 / 32 | 47 / 42 | 72 / 19 | 74 / 17 |
| dose rule: nearest level | 30 / 60 | 53 / 35 | 46 / 42 | 31 / 58 | 53 / 36 | 53 / 35 |
| dose rule: level below | 44 / 43 | 57 / 32 | 56 / 33 | 47 / 41 | 57 / 32 | 57 / 32 |
| dose rule: adverse bracket | 42 / 44 | 63 / 26 | 57 / 32 | 44 / 43 | 63 / 27 | 63 / 26 |
| just-noticeable ratio 1.15 | 32 / 62 | 56 / 37 | 50 / 43 | 34 / 59 | 56 / 38 | 56 / 37 |
| just-noticeable ratio 1.45 | 27 / 56 | 51 / 32 | 45 / 38 | 29 / 54 | 50 / 32 | 51 / 32 |
| detection spread 1 binary step | 31 / 53 | 53 / 34 | 52 / 34 | 41 / 44 | 53 / 34 | 53 / 34 |
| detection spread 3 binary steps | 27 / 64 | 52 / 37 | 36 / 54 | 27 / 63 | 50 / 39 | 52 / 37 |
| detection spread 4 binary steps | 24 / 68 | 50 / 39 | 29 / 63 | 24 / 68 | 45 / 46 | 49 / 40 |
| logistic tail | 27 / 63 | 54 / 35 | 47 / 42 | 30 / 60 | 53 / 35 | 53 / 35 |
| no detection below OAV 1 | 31 / 56 | 54 / 35 | 49 / 39 | 33 / 54 | 53 / 35 | 54 / 35 |
| deterministic at the trained-panel threshold | 30 / 49 | 45 / 43 | 32 / 48 | 30 / 49 | 41 / 45 | 44 / 44 |
| no driving-pressure limit | 18 / 74 | 57 / 32 | 41 / 48 | 20 / 72 | 55 / 33 | 57 / 32 |
| limit 72 mmHg | 31 / 57 | 53 / 35 | 48 / 40 | 33 / 55 | 53 / 35 | 53 / 35 |
| limit 108 mmHg | 28 / 61 | 54 / 34 | 47 / 41 | 30 / 59 | 54 / 35 | 54 / 34 |
| limit 171 mmHg | 24 / 65 | 55 / 33 | 46 / 43 | 27 / 63 | 54 / 34 | 55 / 33 |

- *Driving-pressure limit* (Section 3.1 of the main text): limits slow the passage at fixed volume and orifice; 72 mmHg is the upper end of intrarectal pressure in simulated defecation [S42], 108 and 171 mmHg the peak intra-abdominal pressures of a cough and a jump [S43]. Discarding the passages above the limit instead of slowing them gives sofa 35 / 53, 33 / 55, 33 / 56 and 29 / 60 at 72, 88, 108 and 171 mmHg. No passage changes class: the limit is far above the valve onset.

- *Clothing:* trousers runs exist for the upright sofa and the standing pair only. On those scenes the audible passage is stronger in 91% of doubly smelled pairs with all passages clothed (90% with 70 or 85% clothed), against 80% bare.

- *Distracted observer:* day 51 / 38; the spread of 5 binary steps reverses it (33 / 59).

- *Releaser:* day 44 / 45 (sofa 50 / 39, lift 39 / 48, standing 18 / 72); no selection, so memory equals this comparison.

- *Detection and intensity combined* (the four outcomes of earlier versions, an unsmelled passage counting as the weaker): silent worse / audible worse 0.55 / 4.1 on the sofa, 21.6 / 21.1 in the lift and 0.52 / 3.1 for the day; without the driving-pressure limit 0.52 / 9.2 on the sofa. These shares mostly record which passage is smelled at all.

## S10 Breath sampling

An audible puff reaches the neighbour after a median of 16 s and passes with a time scale of about 3.7 s (10–90%: 1.4–11.5 s). Without a warning, the breathing phase barely matters: for a median passage the worst breath captures 80–99% of the peak. A listener warned by the sound needs about 25 s of breath-hold for an even chance of missing the puff (worst breath below 10% of the peak); 33 s, the typical maximum at end-expiration, gives 63%.

## S11 Origin of the odorants

Two facts the transport model takes as given have chemical explanations in the literature; we did not test them.

*Why sulfur concentration does not follow volume.*

- $H_2S$ comes from sulfate-reducing bacteria [S44] and from cysteine degradation, whose genes are widespread in the gut microbiome [S45]. Sulfate reducers and methanogens both consume $H_2$, but whether they exclude each other is disputed. Sulfidogenic gut contents produced no methane in some studies [S44, S46], whereas faecal sulfide did not differ between methane producers and non-producers in others [S47, S48], $H_2S$ release from incubated faeces did not correlate with methane release ($r = 0.11$) [S49], and methanogenic faeces suppressed sulfide production sixfold in co-incubation [S50].

- The caecal mucosa of rats removes both gases quickly: methanethiol is demethylated to $H_2S$, and $H_2S$ is oxidised mainly to thiosulfate [S51]. Rhodanese is the main detoxifying enzyme in human colon [S52], and SELENBP1 oxidises methanethiol with an apparent $K_m$ of 4.8 nM [S53].

- Only free sulfide can enter the gas, and it is less than 5% of faecal total sulfide (free methanethiol is 22–54% of the total) [S2].

The odorants are thus trace products held at a balance of production and fast consumption, whereas passage volume is set by $N_2$, $CO_2$, $H_2$ and $CH_4$. We infer that this is why sulfur concentration and passage volume are uncorrelated [S7]. Diet moves the balance: a meat-rich

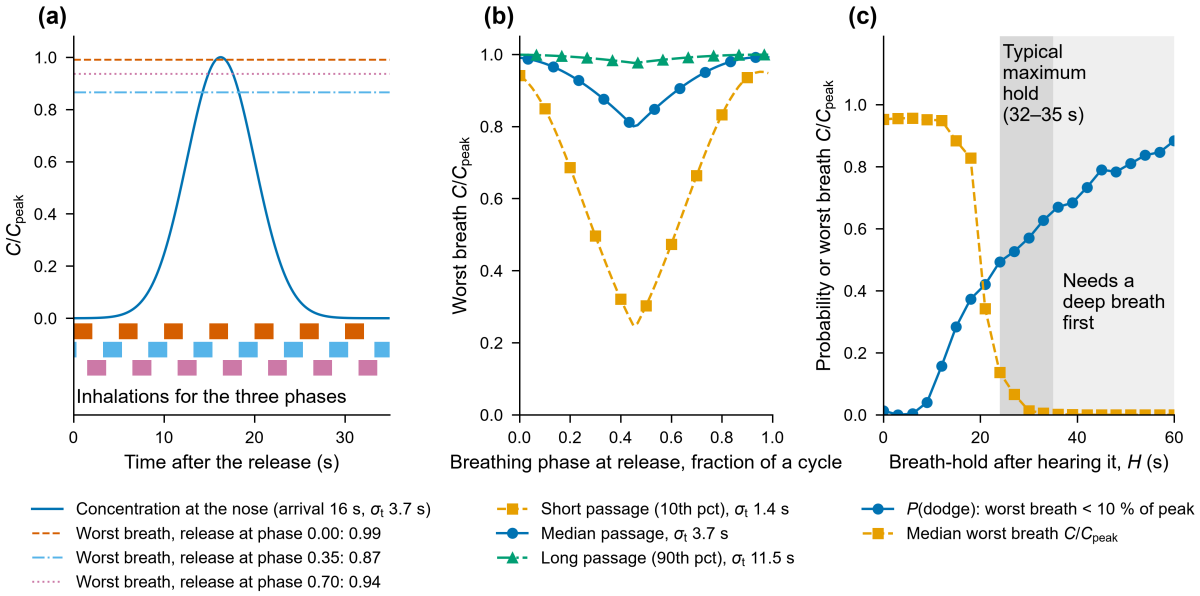



Figure S3: Breath sampling at the neighbour's nose for audible releases. (a) A typical puff (arrival 16 s, passage time scale 3.7 s) and the inhalations of three breathing phases, with the worst breath of each. (b) Without warning: worst breath relative to the peak against the breathing phase at release, for short, median and long passages. (c) Warned: probability of missing the puff (worst breath below 10% of the peak) and median worst breath, against the breath-hold duration after hearing the release (0.5 s reaction latency).

diet raised faecal sulfide from 0.22 to 3.38 mmol $kg^{-1}$ in five volunteers [S54], although faecal sulfide reflects the binding capacity of the stool as much as sulfide production [S55], and the response to animal and plant diets differs between people [S56].

*Diet can make large passages less sulfurous.* Within one diet, sulfur concentration does not change with passage volume (Section S1); across diets it can. In faecal incubations, rapidly fermented fibres produced up to 50 times more gas than poorly fermented fibre while suppressing $H_2S$ by up to 89%, and cysteine raised $H_2S$ 16-fold [S57]. Dietary sulfate tripled sulfide production and cut methane production fivefold, at unchanged stool weight and transit, in three of six subjects [S58]; in the sulfidogenic gut contents of one autopsy subject, $H_2S$ took the place of the $CH_4$ found in a methanogenic one [S46], diverting hydrogen from gas volume to sulfide [S59]. Interventions aimed at sulfur move sulfur alone: bismuth cut faecal $H_2S$ release 23-fold with total gas unchanged [S60], and charcoal, yucca and zinc acetate cut $H_2S$ production by 38–86% without changing gas production in dog faecal incubations, and fed to dogs did not change the number of emissions [S61]. Because audible passages are larger, a diet-driven link of this kind would make audible passages less sulfurous per millilitre, in the direction of the belief. Its size in vivo is unknown: no study has measured sulfur gases and flatus volume together across diets.

*Why methanethiol smells foul.* The human receptor OR2T11 responds to short thiols, including methanethiol, in a copper-dependent way, but did not respond to $H_2S$ in that assay [S62]. Human nasal mucus contains about 2.5 µM copper [S63], below the 30 µM added in the receptor assays, so the receptor's role in vivo is plausible but not shown. Other receptors respond to sulfur ligands without copper [S64–S66]. A receptor that responds to methanethiol and not to $H_2S$ would make methanethiol the characteristic note of flatus independently of thresholds; we infer this, it has not been tested.

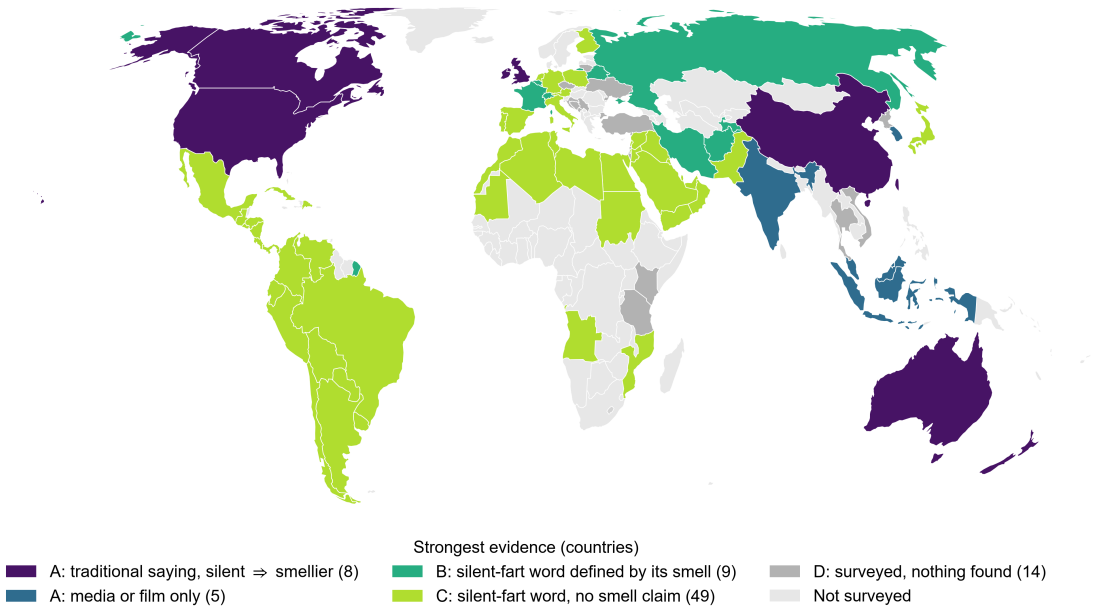



Figure S4: The belief across languages, by evidence level. Each country takes the strongest level among its surveyed languages. A: an explicit saying that silent passages smell worse (traditional, or in media or film only); B: a dedicated word for a silent passage defined by its smell; C: a dedicated word with no smell claim; D: surveyed, nothing found. Documentary presence is not prevalence; historical languages are not mapped. Borders: Natural Earth 1:110m.

## S12 Cross-cultural survey

We searched dictionaries, proverb collections and etymological handbooks in 32 languages or historical stages for a dedicated word for a silent passage and for an explicit claim that it smells worse, and checked every entry used against a retrieved source. Each language is graded by its strongest evidence: A, an explicit saying (traditional, or in media or film only); B, a dedicated word defined by its smell; C, a dedicated word without a smell claim; D, nothing found. The silent/loud distinction is lexicalised in at least five language families. The explicit claim is traditional only in Chinese and English, and smell-defined silent words arose independently at least four times within Indo-European. The survey records whether the belief is documented in a language, not how many speakers hold it.

## S13 Code and data

Code, inputs and processed outputs are in the repository named in the data and code availability statement of the main text, `https://github.com/ccy5123/sbd`; its `README.md` lists the source files of every result. They are not duplicated here.

## S14 Parameters

Table S5: Model parameters: nominal value, range sampled or tested, and basis: taken from a cited measurement, derived from cited data, partly sourced, taken from a non-peer-reviewed source, or assumed. Parameter names are those of the code; the source and notes for every entry are in `params/params.yaml` of the code archive.

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| *Source* | | | | |
| T_gas | 310.15 | 309.85–310.65 | K | measured |
| V_passage | $9.2 \times 10^{-5}$ | $1.7 \times 10^{-5}$–$3.75 \times 10^{-4}$ | $m^3$ | derived |
| V_daily | $7.05 \times 10^{-4}$ | $4.76 \times 10^{-4}$–$1.49 \times 10^{-3}$ | $m^3\,d^{-1}$ | measured |
| n_daily | 10 | 1–20 | $d^{-1}$ | measured |
| d_orifice | 0.004 | 0.001–0.008 | m | partly sourced |
| T_release | 1 | 0.3–20 | s | partly sourced |
| dp_rectal | 2000 | 500–4000 | Pa | assumed |
| dp_max | 11732 | | Pa | measured |
| p_anal_rest | 8666 | 6133–12000 | Pa | measured |
| g_reduced | 1.7 | −0.66 to 4.68 | $m\,s^{-2}$ | derived |
| nu_gas | $1.7 \times 10^{-5}$ | | $m^2\,s^{-1}$ | assumed |
| c_H2S | 1.06 | 0.26–1.86 | $\mu mol\,L^{-1}$ | measured |
| c_CH3SH | 0.21 | 0.05–0.37 | $\mu mol\,L^{-1}$ | measured |
| c_DMS | 0.08 | 0.04–0.12 | $\mu mol\,L^{-1}$ | measured |
| sulfur_sigma_ln | 1.1 | 1–1.25 | – | derived |
| *Odour thresholds* | | | | |
| H2S | 0.41 | 0.41–0.9 | ppbv | measured |
| CH3SH | 0.07 | | ppbv | measured |
| DMS | 3 | | ppbv | measured |
| *Jet and puff* | | | | |
| B_u | 5.8 | 5.8–5.9 | – | measured |
| S_jet | 0.094 | 0.094–0.102 | – | measured |
| K_c | 5.11 | 4.73–5.2 | – | measured |
| Gamma_sj | 2.8 | 2.74–2.86 | – | measured |
| C_puff | 2.6 | 2.54–2.66 | – | measured |
| alpha_th | 0.25 | 0.15–0.35 | – | measured |
| alpha_jet | 0.07 | 0.065–0.08 | – | measured |
| alpha_plume | 0.12 | 0.1–0.16 | – | measured |
| *Environment* | | | | |
| u_handoff | 0.15 | 0.05–0.25 | $m\,s^{-1}$ | assumed |
| u_CBL | 0.17 | 0.11–0.24 | $m\,s^{-1}$ | measured |
| u_CBL_pelvis | 0.1 | 0.05–0.15 | $m\,s^{-1}$ | assumed |
| u_own_eff | 0.11 | 0.07–0.16 | $m\,s^{-1}$ | derived |
| delta_CBL | 0.1 | 0.05–0.15 | m | assumed |
| u_room | 0.08 | 0.03–0.15 | $m\,s^{-1}$ | assumed |
| K_turb | 0.002 | $1 \times 10^{-4}$–0.013 | $m^2\,s^{-1}$ | partly sourced |
| D_H2S_air | $1.65 \times 10^{-5}$ | $1.62 \times 10^{-5}$–$1.77 \times 10^{-5}$ | $m^2\,s^{-1}$ | partly sourced |
| *Geometry* | | | | |
| eye_sit_m | 0.804 | 0.75–0.86 | m | measured |
| eye_sit_f | 0.748 | 0.698–0.799 | m | measured |
| eye_stand_m | 1.64 | 1.536–1.753 | m | derived |
| eye_stand_f | 1.517 | 1.421–1.626 | m | derived |
| L_own | 0.74 | 0.65–0.85 | m | assumed |
| L_neighbour | 1 | 0.8–1.4 | m | assumed |
| *Breathing* | | | | |
| V_tidal | $6.3 \times 10^{-4}$ | $4 \times 10^{-4}$–$8.3 \times 10^{-4}$ | $m^3$ | measured |
| f_breath | 12 | 10–16 | $min^{-1}$ | measured |
| Ti_Ttot | 0.42 | 0.34–0.5 | – | measured |
| T_insp | 2.1 | 1.6–2.7 | s | derived |

Table S5: Model parameters (continued).

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| *Acoustics* | | | | |
| f0_flatus | 160 | 60–300 | Hz | partly sourced |
| bg_library | 40 | 35–45 | dB(A) | assumed |
| bg_living | 35 | 30–45 | dB(A) | assumed |
| bg_office | 45 | 43–55 | dB(A) | measured |
| bg_restaurant | 74 | 67–83 | dB(A) | measured |
| bg_subway | 77 | 74–80 | dB(A) | measured |
| P_th | 1200 | 600–2500 | Pa | partly sourced |
| C_d | 0.85 | 0.6–1 | – | partly sourced |
| alpha_mod | 0.7 | 0.5–1 | – | assumed |
| harm_slope | −12 | −18 to −6 | dB/octave | assumed |
| n_harm | 8 | | – | assumed |
| A_cloth | 1 | 0–3 | dB | partly sourced |
| r_listener | 0.6 | 0.4–3 | m | assumed |
| *Perception and observers* | | | | |
| n_stevens | 0.28 | 0.2–0.45 | – | measured |
| k_w_H2S | 0.95 | 0.8–1.2 | intensity units per decade | partly sourced |
| weber_C | 0.28 | 0.05–0.4 | $\Delta C/C$ | measured |
| tau_adapt | 80 | 40–120 | s | assumed |
| a_inf | 0.35 | 0.25–0.4 | – | partly sourced |
| sniff_ratio | 0.64 | 0.61–0.75 | – | measured |
| t_react | 0.5 | 0.3–1.5 | s | assumed |
| H_max | 33 | 24–90 | s | measured |
| thr_scale_typ | 0.05 | 0.05–0.1 | – | measured |
| thr_sd_bs | 2 | | binary steps | measured |
| distracted_factor | 26 | | – | measured |
| distracted_sd_bs | 3.5 | | binary steps | measured |
| recovery_gain | 1.5 | 1–2 | – | assumed |

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